



SUBMISSION TH914026

The Threshold House

A monsoon-responsive courtyard home on the Meghalaya plateau — where architecture does not occupy the forest, but gently inhabits it.

SITE	SUBMISSION	FORMAT
PARENG · MEGHALAYA	TH914026	A3 LANDSCAPE · 12 SHEETS





01 — Prasanna Chaurasia

ARCHITECTURAL DESIGNER · BIM · COMPUTATION · AI

MA Architecture & Urbanism, Manchester School of Architecture. Earlier qualifications in Civil Engineering and Construction Planning & Management.

Workflows: Revit, Rhino, Grasshopper, Rhino.Inside.Revit, Navisworks, Dynamo, ISO 19650 BIM delivery, AI-assisted design. **Architectural Creative Partner** with xFigura; leads **Urban Matrix** — a multidisciplinary architecture + technology ecosystem. First place, City Futures Challenge (Futurly/xFigura).

PROFILES · [Linktree](#) · [LinkedIn](#) · [Portfolio](#) · [GitHub](#) · [Urban Matrix](#) · [MSA](#)



02 — Ar. Rajvardhan Rangrao Kamire

ARCHITECT & URBAN DESIGNER

MA Architecture & Urbanism, Manchester School of Architecture. Postgraduate project: Tempelhofer Feld, Berlin — neighbourhood development strengthening (not displacing) the social and environmental qualities of place.

Works across scales — architectural detail to neighbourhoods, landscapes and wider urban systems. Registered with the Council of Architecture, India.

PROFILES · [MSA](#) · [LinkedIn](#) · [Instagram](#) · [Behance](#)



03 — Ar. Amandeep Singh Monga

ARCHITECTURAL DESIGNER · URBAN DESIGN & PLANNING

B.Arch — SOA, IPS Academy, Indore (2025). Currently **MSc Urban Design & Planning**, University of Sheffield (2026–27). Trained with Swabhimaan Architects, Ahmedabad.

Research on Jane Jacobs' *Death and Life of Great American Cities* (IJNRD, 2024) — walkability, mixed use, bottom-up planning. Interests: parametric modelling with Rhino, GIS, AI-assisted visualisation.

PROFILES · [LinkedIn](#) · [IJNRD PDF](#) · [Instagram](#)

Pareng, Meghalaya Plateau

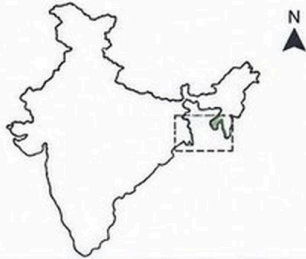
Coordinates	25.5140° N, 91.2521° E
Elevation	1,360 m a.s.l.
Annual rainfall	≈ 11,200 mm
Climate	Tropical monsoon · Köppen Cwa

SITE LEGEND

- 1. Deep 6 m shed roof extends beyond walls for rain protection
- 2. 6 × 6 m open-to-sky courtyard at the heart of the plan
- 3. Perforated copper scupper guides flow into basalt pond
- 4. Verandah & blind shala buffer afternoon and evening humidity
- 5. Six pinstiits — 3 m above saturated grade for flood resilience

01
INDIA

NATIONAL CONTEXT



Location: South Asia
Lat: 8°4' N to 37°6' N
Long: 68°7' E to 97°25' E
Climate: Tropical monsoon
Topography: Plains, Plateaus, Mountains

02
NORTHEAST INDIA

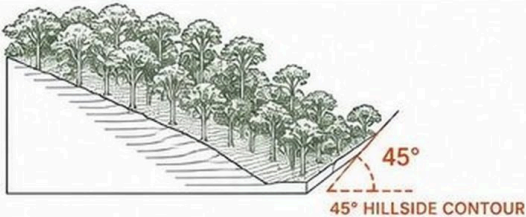
REGIONAL CONTEXT



Annual Rainfall: 10,000–12,000 mm (among the highest in the world)
Monsoon: June – September
Topography: Hills & valleys, altitude 200–2,000 m
Ecology: Biodiversity hotspot

03
MEGHALAYA

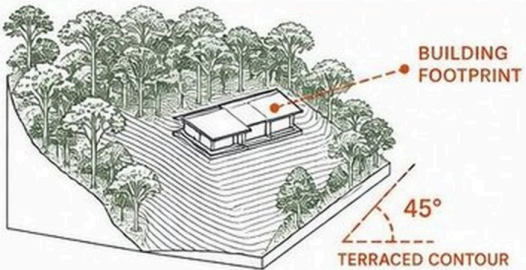
STATE CONTEXT



Landscape: Steep hills, deep valleys
Forest Type: Tropical Wet Evergreen
Rainfall: 10,000–12,000 mm/year
Terrain: 45° typical hillside contour
Settlement Pattern: Dispersed, in harmony with terrain

04
THE THRESHOLD
HOUSE SITE

PROJECT CONTEXT



Setting: Dense rainforest hillside
Contour: ~45° terraced slope
Rainfall: Extreme monsoons
Access: Footpath approach
Site Area: ~[To be determined]
Design Response:
Threshold between forest and dwelling, shelter, adapt, and belong.



THE THRESHOLD HOUSE
Meghalaya rainforest hillside dwelling



EXTREME MONSOONS
10,000–12,000 mm/year



TROPICAL RAINFOREST
Biodiversity hotspot

TEAM: TH914026
COMPETITION:
Archdais The House 3.0

Climate logic

The monsoon is not blocked — it is **choreographed**. South-west winds enter through south verandah louvers, drop into the courtyard rain chain, and exit through northern clerestories. Heat gain deferred; water harvested.

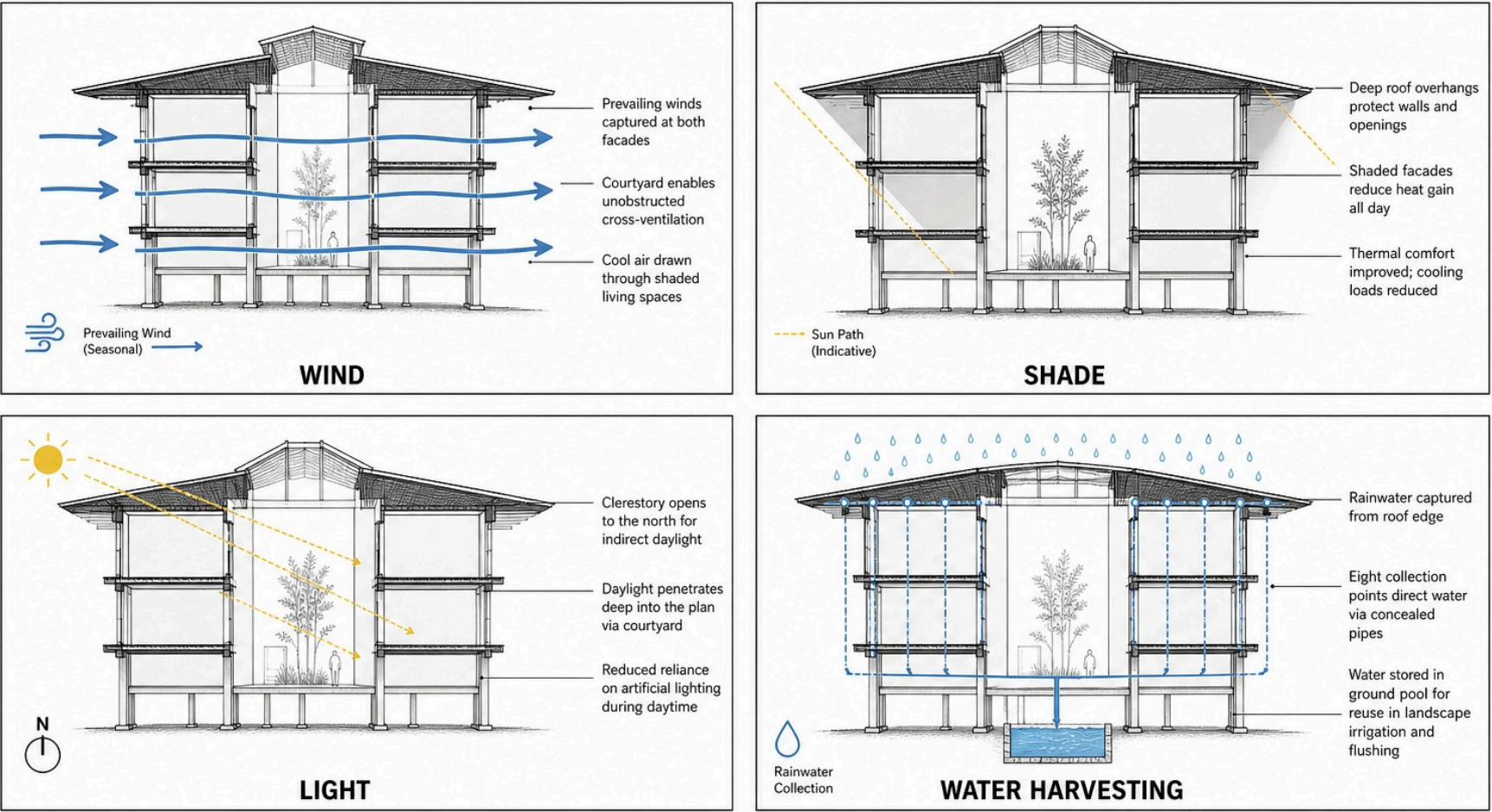
PASSIVE COOLING

- Cross-vent through stilt void
- Thermal mass: basalt pond & laterite walls
- 200 mm RCC slab as night-time heat sink
- Shaded clerestories drawing cool air out

RAIN PATH

- Roof aperture at +9.6 m drops on copper scupper
- 6 m chain & stainless mesh — leaves filtered
- Basalt pond overflows → perocation pit
- Reuse for landscape and flushing

CLIMATE STRATEGY



DESIGN PRINCIPLES



Work with prevailing winds for natural ventilation.



Provide shade and control solar heat gain.



Maximize daylight for comfort and energy efficiency.



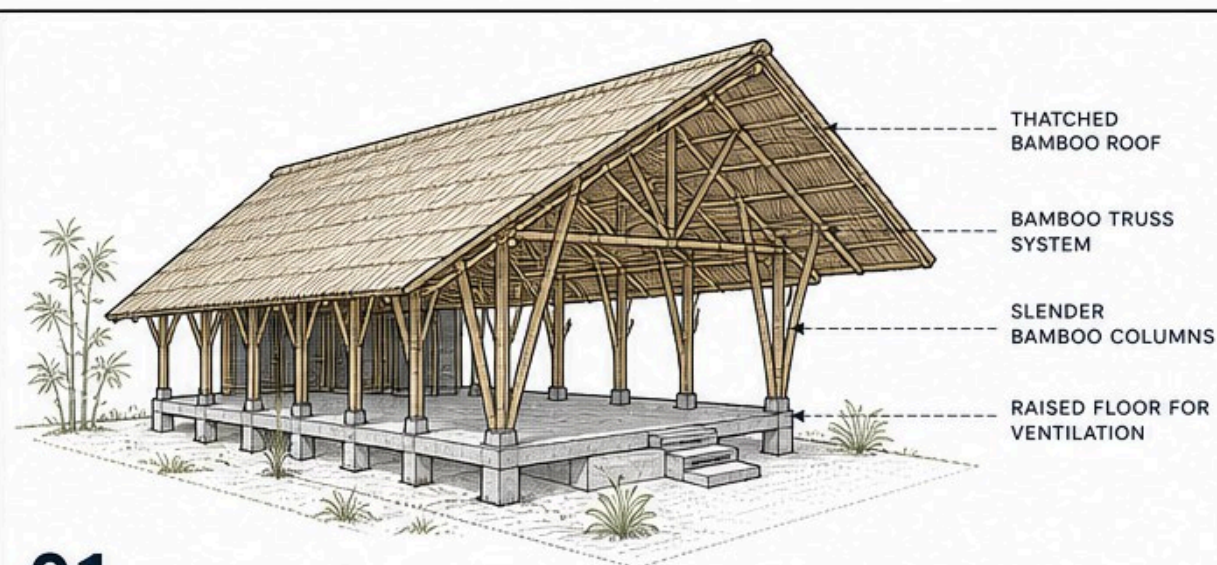
Harvest rainwater for on-site non-potable use.

Lessons applied

FROM BAWA · CORREA · VERNACULAR

PRECEDENTS

THRESHOLD HOUSE | 11



01 RIKSHALLEN

Bamboo hall · Tropical craft

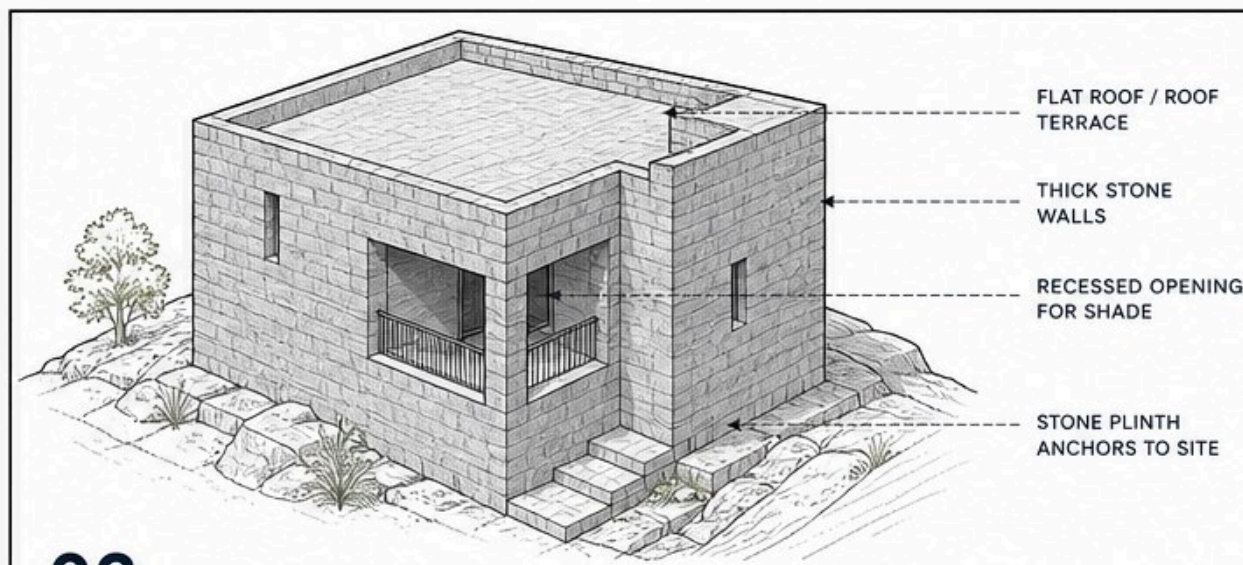
An open community hall that celebrates tropical materials and craftsmanship. Lightweight bamboo structure, generous roof overhang and raised floor create shade, ventilation and adaptability.

THATCHED
BAMBOO ROOF

BAMBOO TRUSS
SYSTEM

SLENDER
BAMBOO COLUMNS

RAISED FLOOR FOR
VENTILATION



02 KIRNAYAKAR

Heavy mass · Stone

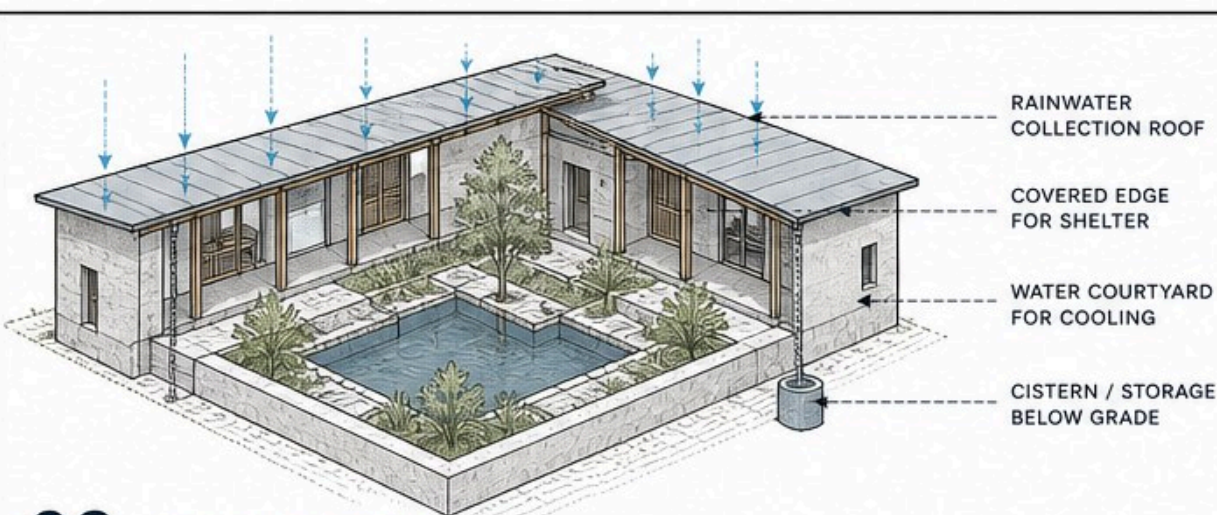
A stone volume that uses mass and permanence to create cool interior spaces. Deep reveals and minimal openings provide shade, privacy and a strong connection to the landscape.

FLAT ROOF / ROOF
TERRACE

THICK STONE
WALLS

RECESSED OPENING
FOR SHADE

STONE PLINTH
ANCHORS TO SITE



03 RAINBARN

Rain courtyard · Water

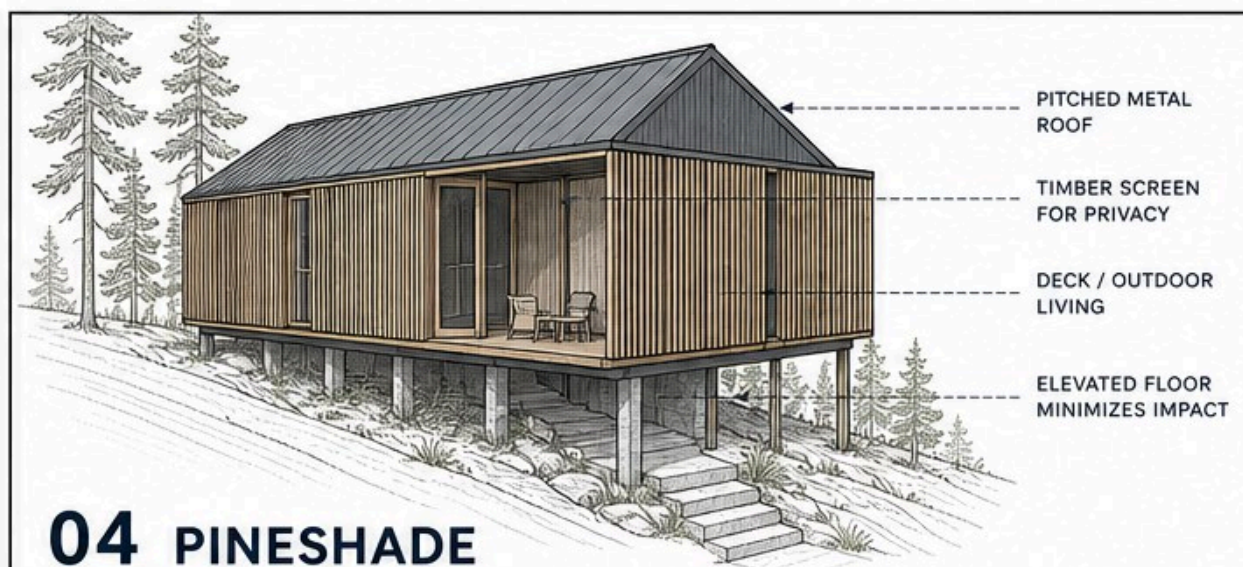
A courtyard house organized around water. Rain is collected, stored and celebrated as a central element that cools the air and supports landscape and daily life.

RAINWATER
COLLECTION ROOF

COVERED EDGE
FOR SHELTER

WATER COURTYARD
FOR COOLING

CISTERN / STORAGE
BELOW GRADE



04 PINESHADE

Timber screens · Retreat

A quiet retreat that balances openness and seclusion. Timber screens filter light and views while an elevated deck embraces the forest and sloping terrain.

PITCHED METAL
ROOF

TIMBER SCREEN
FOR PRIVACY

DECK / OUTDOOR
LIVING

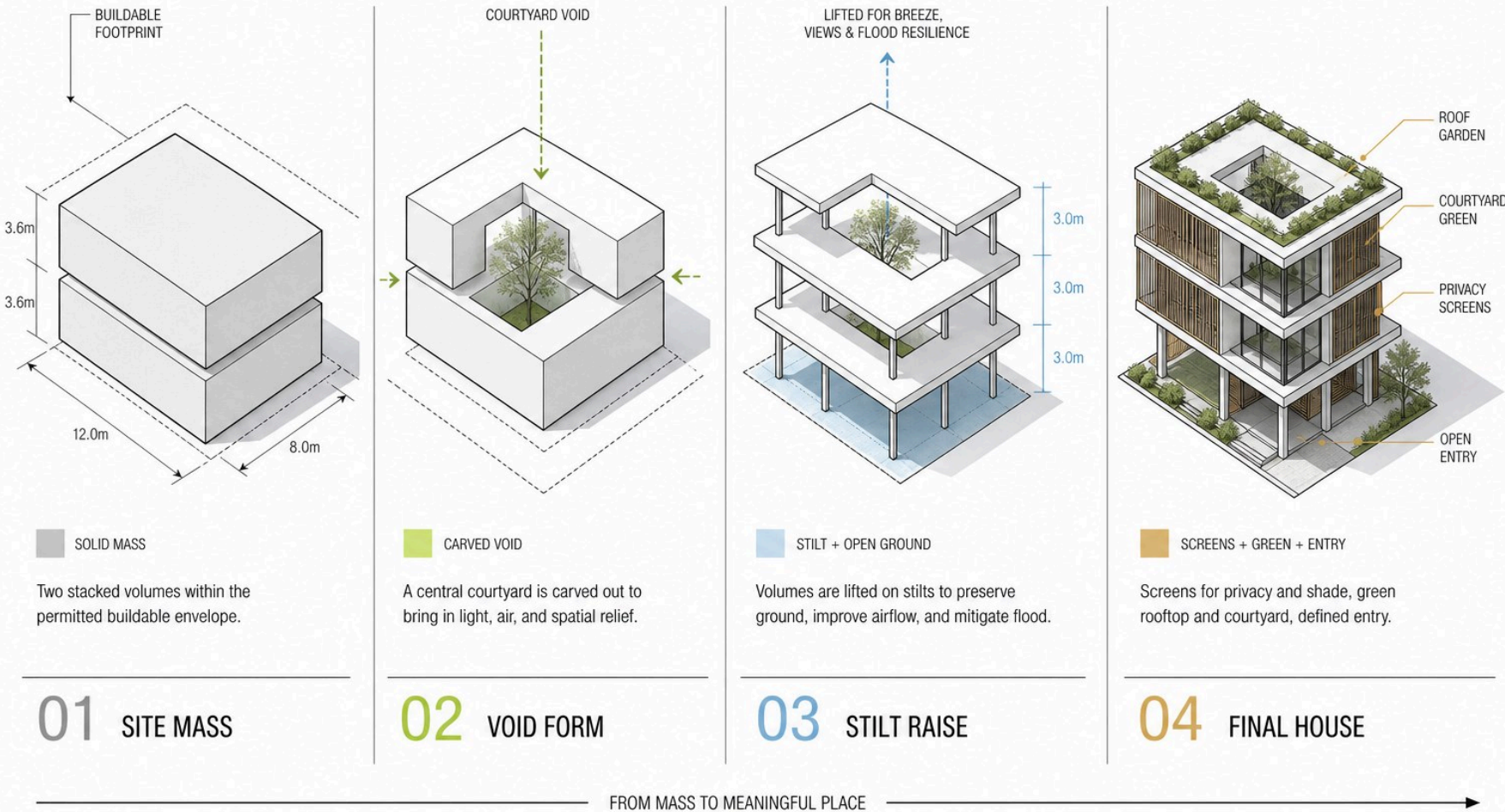
ELEVATED FLOOR
MINIMIZES IMPACT

Sequence to final form

- 1. **Step 01 — Site mass.**
Two stacked volumes 12.2 m × 8.6 m × 3.6 m inside the buildable envelope.
- 2. **Step 02 — Void.**
Central 6 × 6 m courtyard carved out — light, air, spatial relief.
- 3. **Step 03 — Stilt raise.**
Lift volumes 3 m — flood resilience and passive airflow.
- 4. **Step 04 — Final.**
Screens, shades, green rooftop, courtyard, defined entry.
- 5. **Step 05 — Rain chain.**
360 mm copper scupper through stainless mesh into basalt pond.

FORM DEVELOPMENT

THRESHOLD HOUSE | SHEET 09





STILT
60 m²

GROUND
120 m²

FIRST
120 m²

COURTYARD
36 m² open



READING THE FACADE

East faces the courtyard with full-height glazing · south carries the salwood louver and the 6 m cantilever · west is solid masonry with punched openings housing the staircase · north conceals services behind the projecting slab edge and louvred screen.

ELEMENTAL CONCEPT · VISUAL NARRATIVE

RAIN CHAIN

THE POETRY OF RAIN. THE RHYTHM OF DESIGN.

A timeless feature that transforms rainwater into a sensory experience. Our rain chain is more than drainage — it's a sculptural element that connects architecture with nature.

SUSTAINABLE

Natural water flow solution.

SCULPTURAL

Elevates outdoor aesthetics.

SERENE

The sound of rain, reimagined.

DETAILED SIDE ELEVATION

150

300

2700

150

300

- ROOF ASSEMBLY
Metal sheet roofing with insulation and waterproofing
- METAL FASCIA
2mm thick powder coated metal edge flashing
- WOODEN SOFFIT
Teak wood planks with natural oil finish
- RAIN CHAIN HOOK
Stainless steel hook fixed to roof beam
- RAIN CHAIN
Handcrafted copper/brass chain guides rainwater
- VERANDAH LEVEL
FFL +0.90M
- PLANTER BOX
- WATER CHANNEL
Stone lined channel receives water from rain chain
- RCC SLAB EDGE
- RCC SLAB
- RCC COLUMN

RAIN CHAIN MATERIAL & DETAILS

MATERIAL
Copper / Brass

CHAIN LINK
Cup Ø 60mm
Height 60mm
Link Ø 8mm

HOOK
Stainless Steel

FINISH
Natural aged patina

WATER FLOW JOURNEY

Rainwater falls on the roof

Collected at the roof edge

Flows through the rain chain

Directed to the water channel

Stored / Used / Recharged

DESIGNED TO LAST

Corrosion Resistant

Low Maintenance

Eco Friendly

Timeless Beauty

“When it rains, it sings.”
— Architecture in Harmony with Nature

TITLE

Rain Chain

THE POETRY OF RAIN. THE RHYTHM OF DESIGN.

A timeless feature that transforms rainwater into a sensory experience. Our rain chain is more than drainage — it is a sculptural element that connects architecture with nature.

THREE CHARACTERISTICS

1

Sustainable
Natural water flow solution.

2

Sculptural
Elevates outdoor aesthetics.

3

Serene
The sound of rain, reimagined.

01 · ENVIRONMENTAL SECTION

02 · DETAIL ANGLE + SIDE ELEVATION

THE THRESHOLD HOUSE · V34

RAIN CHAIN — ELEMENTAL CONCEPT

SHEET 09 / 12

Vertical Programme

UPPER PRIVATE REFUGE · +4.50 M

Master suite · wardrobe · bath · canopy verandah · teak timber.

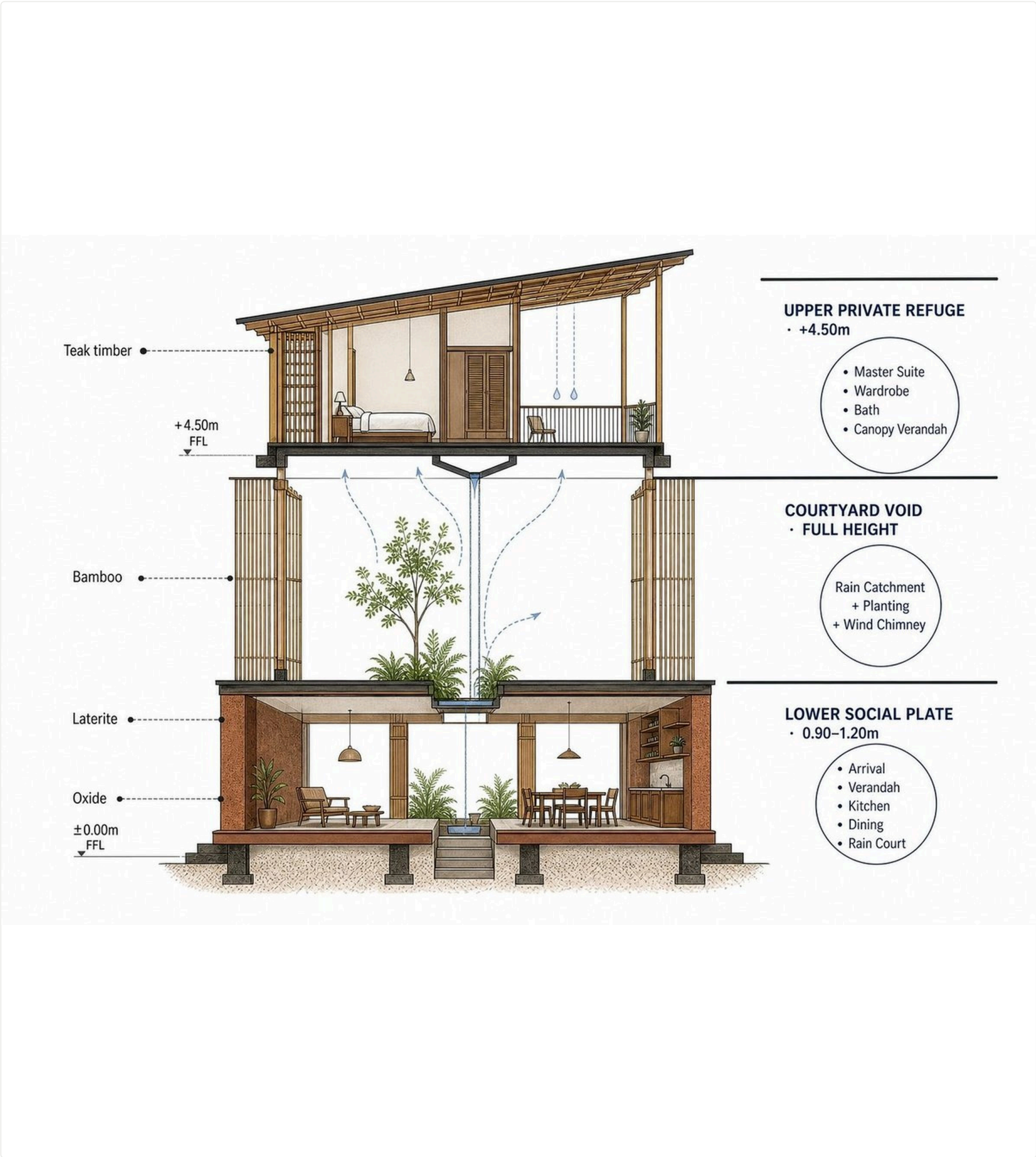
COURTYARD VOID · FULL HEIGHT

Rain catchment · planting · wind chimney · bamboo screens.

LOWER SOCIAL PLATE · ±0.00 M

Arrival · verandah · kitchen · dining · rain court · lateral walls · oxide floors.

STILT 60 m²	GROUND 120 m²	FIRST 120 m²	COURTYARD 36 m²
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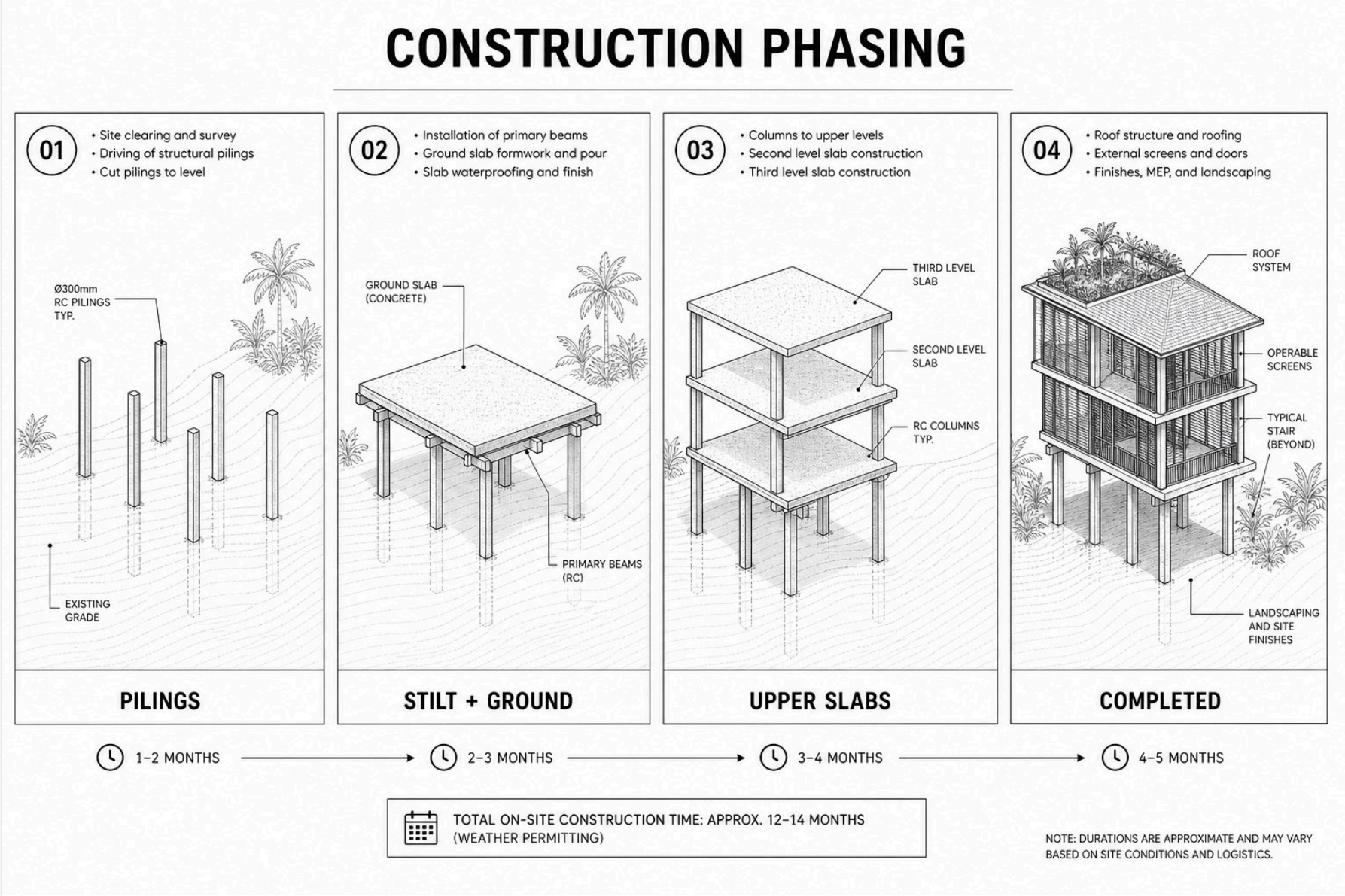


Six phases · ≈ 12–14 months

- **Phase 1 · Pilings** · 4 wk. Six LDK piles (ø 300 mm, 4 m deep) with capping beams.
- **Phase 2 · Ground slab** · 4 wk.
- **Phase 3 · First floor slab** · 4 wk.
- **Phase 4 · Roof slab + waterproofing** · 4 wk.
- **Phase 5 · Fit-out** · 12 wk. Partitions, screens, fixtures, finishes.
- **Phase 6 · MEP + landscaping** · 4 wk.

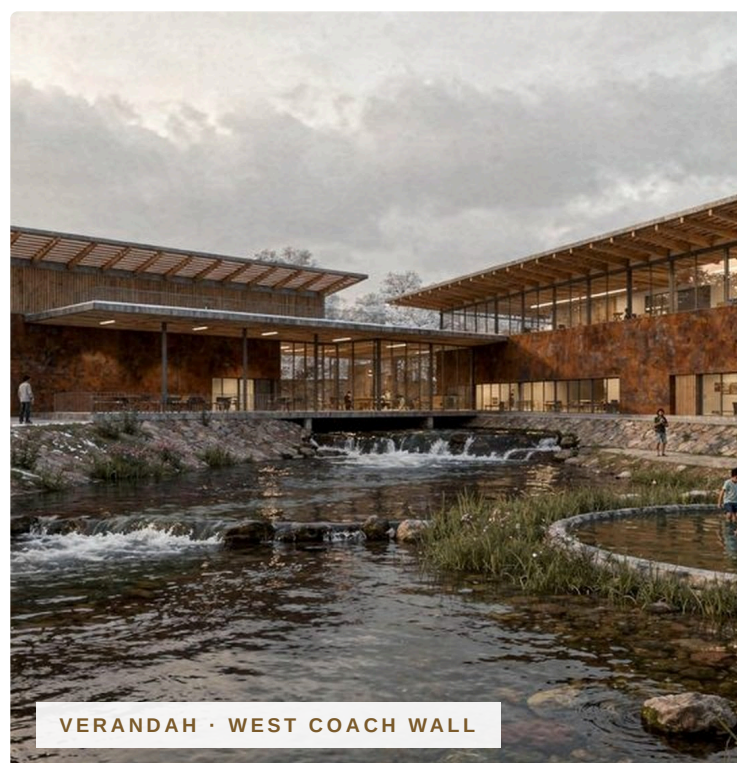
CONSTRUCTION NOTES

Site is remote — access by single forest road after the carriageway junction. Materials chosen for local availability: **salwood**, **teak**, basalt stone, lime plaster, terracotta.



DESIGN REFLECTION

The threshold as an act of inhabiting.



"The Threshold House transforms the boundary between dwelling and forest into an experience — where architecture does not occupy nature, but gently inhabits it."

A monsoon-responsive courtyard home on the Meghalaya plateau. Rain choreographed, wind welcomed, light measured. Three materials do almost all the work: salwood louvers, lime plaster, basalt stone — a finite vocabulary of about 40 components, traceable from renderings to elevations to construction sequencing.

DESIGN LESSONS LEARNED

1. The site teaches the project. Read wind, sun, peak rainfall before any volume study.
2. Choreograph rain; don't block it. Canopy → scupper → chain → mesh → pond.
3. Vertical layering reduces footprint — stilt + void + private refuge.
4. Use what's already there — local limits shape the aesthetic.
5. Draw details that can be built — same hand does render and 1 : 5 detail.

THANK YOU · JURY · ARCHDAIS 2026

*Submitted with gratitude by
Amandeep, Rajvardhan & Prasanna.*

THE THRESHOLD HOUSE

FACADE ANALYSIS + TYPICAL EDGE SECTION

FACADE CONSTRUCTION DETAIL STUDY

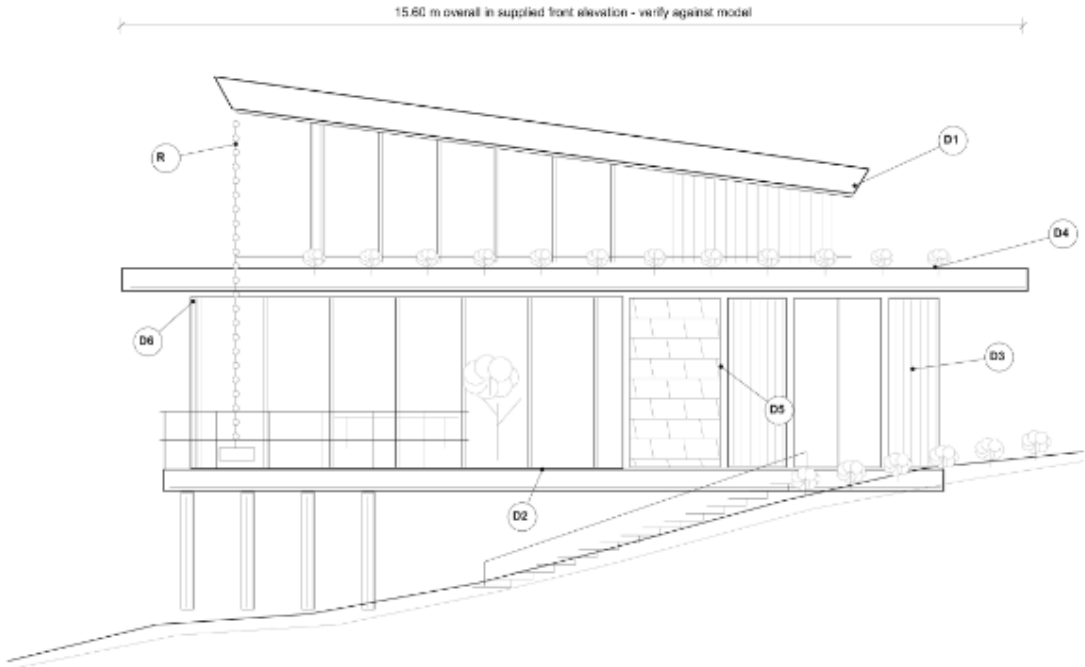
DESIGN DEVELOPMENT / PROPOSED JUNCTIONS

MONOCHROME VECTOR DRAWINGS

01

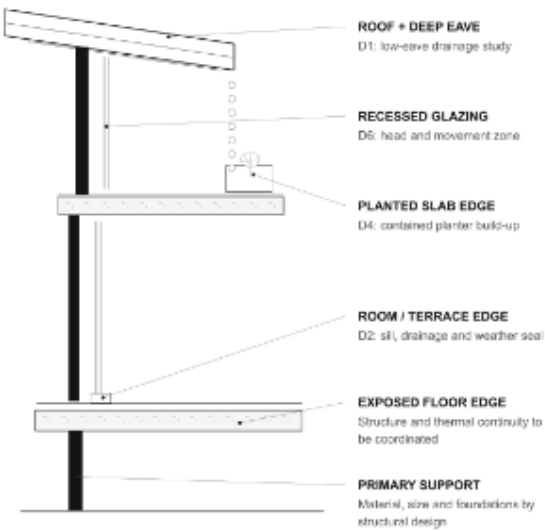
01 / REFERENCE A - FRONT FACADE

Single-pitch rendered reference. Source B is studied on sheet 03. NTS.



02 / TYPICAL VERTICAL EDGE

Proposed build-ups / not a measured building section



REFERENCE GEOMETRY

Retained design cues: single-pitch roof; deep horizontal projections; recessed glazing; timber and stone panels; perimeter planting; elevated supports; external stair.

Source annotations: front width 15.60 m; lateral width 14.60 m; height bands 3.60 / 3.20 / 3.00 m; total 9.80 m. These are image annotations, not surveyed dimensions.

DESIGN COORDINATION

R / The rendered chain sits near the high side of the roof. Confirm roof falls and the concealed connecting drainage route. D1 illustrates a proposed low-eave outlet, not an assumed connection to that chain.

Both supplied sheets are used. Reference B is redrawn on sheet 03 with its roof band, water bay and screen pattern. Differing geometry is shown separately; hidden assemblies remain proposals.

DETAIL INDEX / SHEET 02

- | | |
|----------------------------|--------------------------------|
| D1 Low eave + chain outlet | D2 Glazed sill + terrace drain |
| D3 Timber screen support | D4 Planted slab-edge build-up |
| D5 Stone / glazing jamb | D6 Glazing head + soffit |

DRAWING HIERARCHY

CUT 0.80 PROFILE 0.36 FRAME 0.28 DETAIL 0.18 HATCH 0.10

Nominal plotted weights [mm], at the issued A1 sheet size.

TH / FACADE / 01

NTS. Visible geometry interpreted from image references. Hidden assemblies proposed; sizes and fixings require coordination.

A1 / NOT FOR CONSTRUCTION

Reference A front facade at 15.60 m overall with a typical vertical edge schematic and six detail callouts: D1 low eave + chain outlet, D2 glazed sill + terrace drain, D3 timber screen support, D4 planted slab-edge build-up, D5 stone / glazing jamb, D6 glazing head + soffit. Validation geometry only — not a measured survey.

THE THRESHOLD HOUSE

ENLARGED FACADE JUNCTIONS + CONSTRUCTION INTENT

FACADE CONSTRUCTION DETAIL STUDY

DESIGN DEVELOPMENT / PROPOSED JUNCTIONS

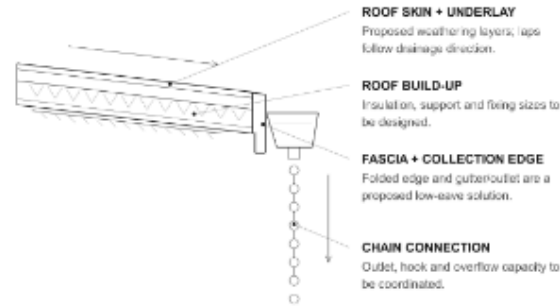
MONOCHROME VECTOR DRAWINGS

02

All six details are newly drawn vectors. They explain proposed assemblies without inventing verified sizes, product profiles or engineering specifications.

D1 LOW EAVE + RAIN CHAIN

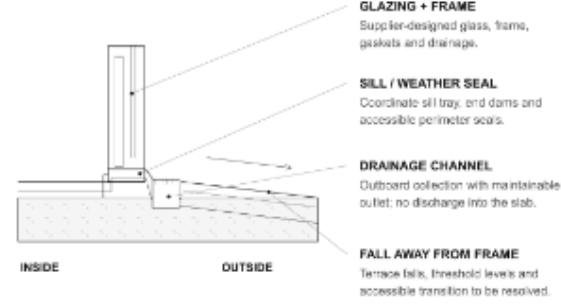
SECTION / PROPOSED DRAINAGE OPTION



Keep the single-pitch silhouette and deep eave. Outlet position, roof fall, overflow, loads and chain restraint require project coordination.

D2 GLAZED SILL + TERRACE

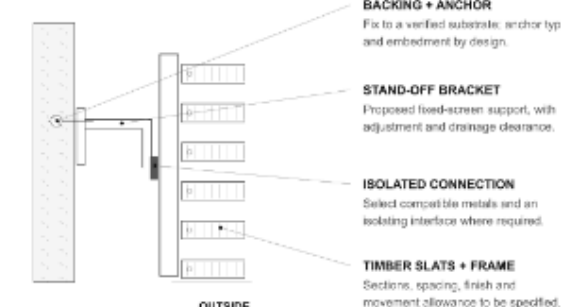
SECTION / PROPOSED WEATHERING STRATEGY



Establish actual internal/external levels before detailing. Frame profile, sill depth, channel size and waterproofing laps are not dimensioned from an image.

D3 TIMBER SCREEN SUPPORT

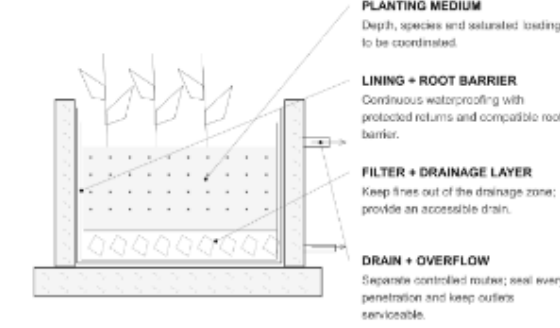
VERTICAL SECTION / FIXED-SCREEN OPTION



The visible screen language is retained. Fixed support is proposed; operability is not established by the references. Bracket and anchor dimensions remain TBC.

D4 PLANTED SLAB EDGE

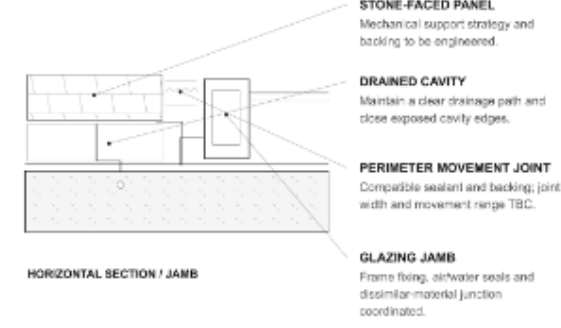
SECTION / PROPOSED CONTAINED PLANTER



Coordinate saturated soil load, slab capacity, lining continuity, overflow destination and maintenance access. Do not drain into the facade cavity.

D5 STONE / GLAZING JAMB

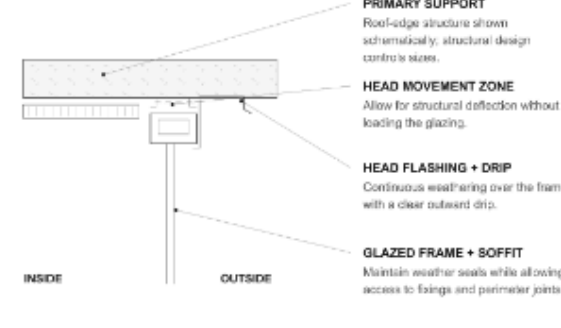
PLAN DETAIL / PROPOSED CAVITY JUNCTION



Stone thickness, backing wall, supports, cavity barriers and movement joints depend on the selected system and project requirements.

D6 GLAZING HEAD + SOFFIT

SECTION / PROPOSED HEAD JUNCTION



Frame head, deflection allowance, flashing, support and soffit fixing must be resolved together. No structural dimensions are implied by this diagram.

TH / FACADE / 02

NTS. Visible geometry interpreted from image references. Hidden assemblies proposed; sizes and fixings require coordination.

A1 / NOT FOR CONSTRUCTION

Six enlarged junctions showing sections and proposed drainage / weather strategies; each callout explains the assumed build-up. Figure geometry referenced from supplied context. NTS — design-development intent, not yet ratified for fabrication.

THE THRESHOLD HOUSE
REFERENCE B FACADES + CENTRAL WATER-BAY ENLARGEMENT

FACADE CONSTRUCTION DETAIL STUDY
DESIGN DEVELOPMENT / PROPOSED JUNCTIONS
MONOCHROME VECTOR DRAWINGS

03

The four SVG elevations form a separate facade study. Their roof-band and panel arrangements are not forced onto reference A.

NORTH ELEVATION / REFERENCE B



Facade geometry interpreted from the supplied SVG / NTS

SOUTH ELEVATION / REFERENCE B



Facade geometry interpreted from the supplied SVG / NTS

EAST ELEVATION / REFERENCE B



Facade geometry interpreted from the supplied SVG / NTS

WEST ELEVATION / REFERENCE B



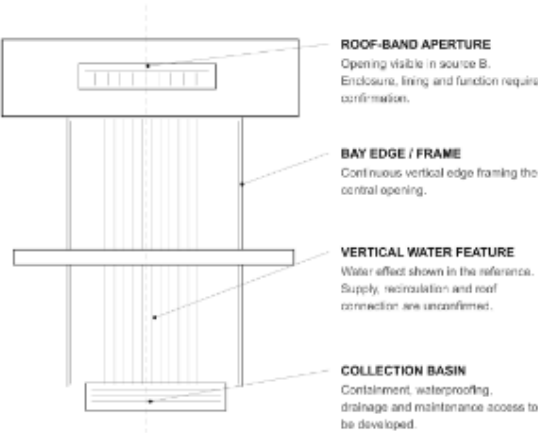
Facade geometry interpreted from the supplied SVG / NTS

FACADE STUDIES, NOT MEASURED SURVEYS

Both source families inform this package. Matching the images does not establish hidden construction, actual dimensions, structural capacity or waterproofing performance.

D7 / ROOF OPENING + WATER BAY

ENLARGED ELEVATION / VISIBLE RELATIONSHIPS



USING BOTH REFERENCES

A: sloping roof, deep eaves, recessed rooms, planted terrace and exposed supports. B: roof-band aperture, central water bay, salted panels and distinct north / south / east / west patterns.

D2, D3, D5 and D6 describe common facade junction types, subject to the selected geometry. D7 is a sloping-roof option for A. D7 preserves the opening and water-bay relationship visible in B.

Confirm a coordinated model before setting out: roof configuration, opening widths, mullion positions, floor alignment and water-system connections differ between the images.

Reference B reinterpreted with the roof-band aperture, central water bay, salted panels and distinct north / south / east / west patterns. D7 enlarges the opening + water-bay relationship. Geometry interpolated from supplied reference — NTS.